Problem Pack

Directions for the PDE BVP Project (Weeks 5–9)

Each direction gives: the phenomenon, the mathematical model, the numerical challenges, the specific things your report should address, and a sanity check you can construct yourself — no reference solution is handed out. Direction 3 can be completed with **either FDM or FEM** — choose one and justify the choice (see the brief). The disk and mixed-boundary directions instead prescribe the finite-difference formulations stated below (a Cartesian cut-cell closure; two FDM boundary closures). Teams who choose FEM should therefore take Direction 3 — or propose their own elliptic BVP in your Week-5 milestone entry. The analytic test-problem bank (`03_Scaffolding/test_problem_bank.py`) contains independent exact solutions for the *rectangular* core problems; the disk direction has no bank entry — building its mask and boundary rule is part of the work.

You may choose one of these directions, or propose your own elliptic BVP (state it in your Week-5 milestone entry).

Write the one-page problem definition sheet (`06_Templates/problem_definition_sheet.tex`) before coding: Part 1 (the continuous problem, sign conventions, method choice) is due with your Week-5 issue ticket; **Part 2** (the discrete operator — stencil for FDM, weak form for FEM) is due with Part 1 for FDM teams and after Lecture 4 for FEM teams. Then, before every new solve, complete the pre-computation protocol (predictions first); and if an AI writes solver code, it must pass the five-gate audit before its output is used — both protocols are at the end of this pack.

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1. Manufactured Poisson problem on a disk

Phenomenon. Verification by construction — the course-wide method for checking a solver with *no* reference solution — but now on a *curved* boundary, where the mask/index-map machinery and the geometry of the boundary treatment become part of the method.

Model. The Poisson problem on the unit disk,

$$-\Delta u = f \quad \text{in } \Omega, \qquad \Omega = \{(x, y) : x^2 + y^2 < 1\}, \qquad u = g \quad \text{on } \partial\Omega.$$

For this direction, use Cartesian finite differences with the prescribed Shortley–Weller closure; a polar-coordinate comparison remains optional. A good manufactured solution is the harmonic exponential

$$u_* = e^x \cos y \quad \Rightarrow \quad f = 0,$$

so the forcing vanishes and the whole numerical challenge is the boundary data $g = e^x \cos y$ on the circle. (Any harmonic u_* keeps $f = 0$; $u_* = r^4 \cos 4\theta$ is another option.) This boundary-focused test does not exercise nonzero forcing assembly. Include the companion manufactured solution

$$u_\dagger = e^x \cos y + 1 - x^2 - y^2, \qquad f_\dagger = 4, \qquad g_\dagger = e^x \cos y,$$

which has the same boundary data but checks the interior right-hand side.

Numerical challenges.

- Embed the disk in a uniform Cartesian grid on $[-1, 1]^2$, retain the active nodes $P = (x_i, y_j)$ with $x_i^2 + y_j^2 < 1$, and build the index map $(i, j) \mapsto k$. Grid nodes outside the disk are never unknowns. See `domain_mask_skeleton.py` and the non-rectangular-domains guide.
- Use the prescribed **Shortley–Weller (cut-cell) closure** below. Do not replace the circle by a staircase and do not assign boundary values at an outside grid node: evaluate g at the exact intersection of each cut grid line with the circle.

- Rows adjacent to a cut edge have unequal directional distances and are generally nonsymmetric after assembly. Record the asymmetry and check the sign pattern and row sums; do not claim that the matrix is symmetric positive definite. Very small cut distances create large coefficients and may worsen conditioning.
- Separate discretisation error from algebraic error. On every mesh, make the linear-solver error negligible relative to the measured solution error and report both the residual and stopping tolerance.

Prescribed Shortley–Weller/cut-cell row. For an active node P, let d_E, d_W, d_N, d_S be the distances from P in the four coordinate directions. A distance equals h when the neighbouring grid node is active. If that neighbour is outside, the distance is instead the positive distance from P to the exact circle intersection and lies in $(0, h]$ (equality is possible when a grid node lies exactly on the circle). For a unit coordinate direction e, compute it from

$$d(P, e) = -P \cdot e + \sqrt{(P \cdot e)^2 + 1 - \|P\|_2^2}.$$

At a cut point $B = P + d(P, e)e$, use the known value $u(B) = g(B)$.

Approximate the two directional second derivatives by the unequal-spacing three-point formulas

$$D_{xx}^{\mathrm{SW}} u(P) = \frac{2}{d_E + d_W}\left(\frac{u_E - u_P}{d_E} - \frac{u_P - u_W}{d_W}\right), \qquad D_{yy}^{\mathrm{SW}} u(P) = \frac{2}{d_N + d_S}\left(\frac{u_N - u_P}{d_N} - \frac{u_P - u_S}{d_S}\right),$$

and impose

$$-D_{xx}^{\mathrm{SW}} u(P) - D_{yy}^{\mathrm{SW}} u(P) = f(P).$$

Here u_E, for example, means an interior unknown when the east grid node is active and means $g(P + d_E e_x)$ when the east edge is cut. Move every known boundary term to the right-hand side with its actual coefficient; for a cut east edge the contribution is

$$\frac{2}{(d_E + d_W)d_E}\, g(P + d_E e_x),$$

not g/h^2. When all four distances equal h, this row reduces exactly to the standard five-point stencil for $-\Delta$.

The unequal-spacing second derivative has an $O(h)$ local truncation term at an irregular node when the opposing distances differ. Because such nodes are confined to a thin boundary layer, the classical Shortley–Weller solution is often second-order accurate in the nodal maximum norm for smooth Dirichlet problems on smooth domains, despite those locally first-order rows. Treat this as a convergence claim to test, not as permission to assume a slope in advance; derivatives of the numerical solution can converge more slowly.

Optional contrast: mask (general) vs. polar coordinates (specialised). The Cartesian cut-cell construction above extends to many implicitly described domains, provided grid-line intersections are located robustly. The disk has special structure you could exploit instead: write the Laplacian in polar coordinates (r, θ),

$$\Delta u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta},$$

discretise on a grid $r_i = ih$, $\theta_j = j\Delta\theta$ with the *periodic* boundary in θ and $u = g$ at $r = 1$. All angular values at $r = 0$ represent the same physical point, so use one centre unknown rather than duplicating it. One consistent origin equation follows from the circular mean identity for a smooth function,

$$\Delta u(0) = \frac{4}{h^2}\left(\frac{1}{N_\theta}\sum_{j=0}^{N_\theta - 1} u(h, \theta_j) - u(0)\right) + O(h^2),$$

with enough uniformly spaced angles for the quadrature error to be smaller than the radial error. Refine $\Delta\theta$ together with h and state the coupling used. A useful optional comparison is the observed error and work of the Cartesian Shortley–Weller method versus this disk-specific polar method.

Your report should address:

- The active/inactive mask (a plot or sketch), the index map, and the matrix structure (spy plot, sign pattern, nonsymmetry, and conditioning trend under refinement).
- The Shortley–Weller row written out for one node with a cut edge: report the intersection point, all four distances, matrix coefficients, and boundary contribution to the right-hand side.
- A four-mesh refinement table $(h, h/2, h/4, h/8)$ in a *named* discrete L^2 norm and the nodal L^∞ norm, with fitted observed orders and the residual/tolerance used at each level. Define the norm weights and use the same family of Cartesian grids throughout. Run the full table for u_* and at least two levels for $u_\dagger$ to confirm that nonzero forcing is assembled correctly.
- A consistency check that the computed cut points lie on the circle, $0 < d \le h$ on every cut edge, and the SW row reduces to the five-point row whenever $d_E = d_W = d_N = d_S = h$.
- What limits the rate: irregular boundary rows, small cut cells, round-off, or the solver tolerance — and how you separated them.

Sanity check. With $u_* = e^x \cos(y)$ and the Shortley–Weller closure, the solution error should decrease consistently under refinement and normally approach second order in both requested nodal norms. A coarse-grid slope is not a proof: inspect the last three levels and also monitor the smallest cut distance. Do *not* use $u_* = x^2 - y^2$ for the convergence study: both the five-point interior stencil and the unequal-spacing boundary formula are exact on this quadratic, so the nodal error can remain at round-off. More generally, the five-point interior stencil is exact on harmonic polynomials of degree less than four; a cubic can still expose the cut-cell closure, but is not a strong test of the interior truncation error. The algebraic residual $\|F - A_h u_h\|$ is not the discretisation error; normalise it, state its norm, and report it separately.

2. Mixed boundary data (Dirichlet / Neumann / Robin)

Phenomenon. Real problems do not come with Dirichlet data everywhere. Heat exchangers and electrostatic devices mix prescribed values, imposed fluxes, and convective or radiative exchange. Pressure-Poisson equations also lead to Neumann data, but those data come from the momentum or projection equations and are not generically zero. The boundary closure is part of the numerical method.

Reproducible model. Let $\Omega = (0,1)^2$ and use

$$-\Delta u = f \quad \text{in } \Omega, \qquad u = g \quad \text{on } \Gamma_D,$$

where Γ_D consists of $x = 0$, $y = 0$, and $y = 1$, including all four corners. On the open right edge $\Gamma_B = \{1\} \times (0,1)$, run two variants:

$$\text{Neumann: } \partial_n u = q, \qquad \text{Robin: } \partial_n u + \beta u = r, \quad \beta = 2.$$

Use the manufactured solution

$$u_*(x,y) = e^{x+y}, \qquad f = -2e^{x+y},$$

with

$$g = u_*|_{\Gamma_D}, \qquad q(y) = e^{1+y}, \qquad r(y) = 3e^{1+y}.$$

The outward normal on $x = 1$ is $(1,0)$, so $\partial_n u_* = u_x(1,y)$. Giving the Dirichlet edges ownership of the corners removes any ambiguity about which discrete row is assembled there.

Two required boundary closures. Use a nodal grid $x_i = ih$, $y_j = jh$, $h = 1/N$. Dirichlet values are eliminated or imposed strongly. At a right-edge unknown $U_{N,j}$, $1 \le j \le N-1$, compare:

1. **First-order boundary row:**
$$\frac{U_{N,j} - U_{N-1,j}}{h} = q_j$$
for Neumann data, or
$$\frac{U_{N,j} - U_{N-1,j}}{h} + \beta U_{N,j} = r_j$$
for Robin data. This row replaces the PDE row at that boundary node.

2. **Second-order ghost-point closure:** introduce a conceptual value $U_{N+1,j}$ and use the centred boundary condition
$$\frac{U_{N+1,j} - U_{N-1,j}}{2h} = q_j$$
or
$$\frac{U_{N+1,j} - U_{N-1,j}}{2h} + \beta U_{N,j} = r_j.$$
Eliminate the ghost value and impose the five-point equation for $-\Delta u = f$ at $(1, y_j)$. Thus
$$U_{N+1,j} = U_{N-1,j} + 2hq_j$$
in the Neumann case and
$$U_{N+1,j} = U_{N-1,j} + 2h(r_j - \beta U_{N,j})$$
in the Robin case. No ghost value remains in the linear system.

The two assemblies are generally nonsymmetric: derivative rows have different scaling from interior PDE rows, and ghost elimination doubles the inward coupling in a boundary PDE row. Do not use conjugate gradients merely because the continuous Laplacian is self-adjoint. Use a sparse direct solver or justify a nonsymmetric Krylov method and preconditioner. Because PDE rows scale like h^{-2} while first-order boundary rows scale like h^{-1}, define a scaled or componentwise residual before comparing residuals across meshes.

This comparison changes more than the formal order of a derivative formula. The first closure replaces the boundary PDE row by a derivative row, whereas the ghost closure enforces the PDE at the boundary after eliminating an auxiliary value. State this formulation difference, scale rows before comparing residuals, and do not attribute every observed difference solely to "first versus second order."

Solvability and the pressure-Poisson interpretation.

This manufactured problem has a nonempty Dirichlet boundary, so it requires no integral compatibility condition. For a pure-Neumann problem with the sign convention above and $q = \partial_n u$, solvability requires
$$\int_\Omega f \, d\Omega + \int_{\partial\Omega} q \, ds = 0,$$
and the solution is determined only up to a constant. Fix its mean or one nodal value after enforcing a compatible discrete right-hand side. In a pressure-Poisson problem, derive q from the velocity/momentum discretisation; setting $p = 0$ along an entire edge is a physical Dirichlet condition, not just a pressure-gauge choice. A Robin portion with positive β removes the constant nullspace and does not require the pure-Neumann compatibility condition.

Your report should address:

- Write the first-order and ghost-eliminated right-edge rows by hand, including the signs and all factors of h. Verify each row by substituting sampled values of u_*.

- For both the Neumann and Robin variants, give four-mesh error tables for both closures in a named discrete L^2 norm and the nodal L^∞ norm. Fit rates only after identifying the asymptotic range.
- Report errors separately on the right boundary and in the interior, together with a scaled algebraic residual. Explain whether the boundary closure limits the global error.
- Show a matrix spy plot and quantify $\|A - A^T\|/\|A\|$. State why your chosen solver is appropriate for that matrix.
- *Optional depth:* repeat with pure Neumann data, check the discrete compatibility defect, impose a gauge, and demonstrate the constant null vector before the gauge is applied.

Sanity check. The first-order closure should approach order one in the boundary L^∞ error. The ghost-point closure should approach order two for this smooth solution in both requested global norms. These are asymptotic expectations, not acceptance based on one mesh pair. Confirm the outward-normal sign using $q = e^{1+y} > 0$ and ensure that the four Dirichlet corners are never overwritten by a Neumann or Robin row. A small algebraic residual alone does not validate the boundary discretisation.

3. Variable diffusion: $-\nabla \cdot (a(x,y)\nabla u) = f$

Phenomenon. Steady heat in a heterogeneous medium — layered composites, soil, tissue — where the conductivity a varies in space and may jump across material interfaces. The physically correct operator is the *divergence form* $-\nabla\cdot(a\nabla u)$, which guarantees that the normal flux $-a\,\partial_n u$ is continuous across interfaces.

Model.

$$-\nabla \cdot \big(a(x,y)\,\nabla u\big) = f \quad \text{in } \Omega, \qquad u = g \text{ on } \partial\Omega, \qquad 0 < a_{\min} \le a(x,y) \le a_{\max} < \infty.$$

Smooth manufactured benchmark. On $\Omega = (0,1)^2$, take

$$a(x,y) = 1 + x, \qquad u_*(x,y) = \sin(\pi x)\sin(\pi y), \qquad g = 0,$$

and derive

$$f(x,y) = 2\pi^2(1+x)\sin(\pi x)\sin(\pi y) - \pi\cos(\pi x)\sin(\pi y).$$

This case checks the $\nabla a \cdot \nabla u$ contribution without forming that term separately.

Layered-interface benchmark. On the same square, let

$$a(x) = \begin{cases} 1, & x < \frac{1}{2}, \\ 10, & x > \frac{1}{2}, \end{cases} \qquad f = 0,$$

and prescribe $g = u_*$ on the whole boundary, where

$$u_*(x) = \begin{cases} \dfrac{20}{11}x, & 0 \le x \le \frac{1}{2}, \\ \dfrac{2x+9}{11}, & \frac{1}{2} \le x \le 1. \end{cases}$$

Then u_* is continuous, its derivative jumps, and the physical flux $-au_x = -20/11$ is continuous. For the Cartesian FDM experiment use odd numbers of intervals, for example $N = 31, 63, 127, 255$, so $x = 1/2$ lies midway between two grid columns rather than on a node.

Conservative discretisation. For FDM, use one shared positive coefficient on each grid face and assemble

$$(A_h U)_{i,j} = \frac{1}{h^2}\big[a_E(U_{i,j} - U_{i+1,j}) + a_W(U_{i,j} - U_{i-1,j}) + a_N(U_{i,j} - U_{i,j+1}) + a_S(U_{i,j} - U_{i,j-1})\big].$$

For smooth a, the arithmetic midpoint approximation $a_E = (a_{i,j} + a_{i+1,j})/2$ is second-order accurate. At a material interface, use the resistance or harmonic average. When the interface bisects the grid edge,

$$a_E = \frac{2a_L a_R}{a_L + a_R};$$

more generally, if distances $d_L + d_R = h$ of the edge lie in the two media, use $a_E = h/(d_L/a_L + d_R/a_R)$. Arithmetic averaging remains positive and can still produce a symmetric matrix, but it generally gives the wrong interface resistance and flux. Symmetry comes from sharing the *same* face coefficient between the two adjacent rows, not from choosing an arithmetic average.

FEM teams should use the weak form

$$\int_\Omega a\nabla u_h \cdot \nabla v_h \, dx = \int_\Omega f v_h \, dx,$$

with strong Dirichlet data. Align the layered mesh with $x = 1/2$ and integrate a separately on each material region; an element must not be assigned a single averaged coefficient while it straddles an unresolved jump.

After eliminating Dirichlet degrees of freedom, either conservative assembly is symmetric positive definite. Replacing Dirichlet rows by identity while leaving their columns unchanged destroys matrix symmetry. With positive shared face coefficients, the Cartesian FDM matrix is also an M-matrix and satisfies a discrete maximum principle; coefficient contrast worsens conditioning but does not by itself invalidate that property. A pointwise discretisation of $-a\Delta u - \nabla a \cdot \nabla u$ is inappropriate at a discontinuity and may lose conservation, symmetry, and the monotone sign pattern.

Your report should address:

- Write the conservative FDM face-flux stencil or FEM weak form and state exactly how coefficients and Dirichlet data are assembled. Report $\|A - A^T\|/\|A\|$ and assess positive definiteness. The primary evidence is the derivation after Dirichlet elimination with positive shared face coefficients. Acceptable numerical corroboration is a sparse Cholesky factorisation (where available) or a smallest-eigenvalue estimate with `eigsh`; random Rayleigh quotients are diagnostics, not proofs.
- For the smooth benchmark, give four-mesh L^2 and L^∞ error tables, fitted asymptotic orders, and algebraic residuals. Confirm the manufactured forcing independently before using it to test the solver.
- For the layered benchmark, compare arithmetic and harmonic/resistance face coefficients in FDM, or compare an interface-fitted FEM mesh with an explicitly documented unresolved-interface treatment. Plot u and the numerical normal flux across $x = 1/2$ and quantify both the solution error and the flux jump.
- Show that the layered numerical solution stays between its boundary extrema. Relate this observation to the matrix sign pattern rather than treating it as evidence of accuracy.
- Compare conditioning or iterative-solver iterations under mesh refinement and under increasing contrast a_R/a_L; keep the residual tolerance comparable across cases.

Sanity check. The smooth problem should approach second order for the specified FDM stencil and for standard piecewise-linear FEM in L^2 on a regular mesh. In the aligned layered test, the harmonic/resistance flux should be constant to the accuracy of the linear solve and the piecewise-linear exact solution should be reproduced at the nodes up to round-off. Arithmetic averaging at the jump is a deliberate comparison, not an acceptable reference closure. A symmetric matrix alone does not prove that the interface physics is correct.

4. Depth directions (★★: add one substantial comparison)

Depth work sits at the ★★ level of the brief: a team working at the basic level may stop at the core content, while teams aiming higher add exactly one of the following to their core direction. Direction 2 already requires comparing two boundary closures — that required comparison counts as the depth item for that direction, so those teams need not add another comparison from the list below. Having the item is not itself a Dimension B score: the rubric levels that dimension by quality, so the level reached depends on how well the required comparison is executed and argued. Deepening further (cost-versus-accuracy, a third variant, or the pure-Neumann depth inside that direction) is the optional route to the higher levels and must not simply repeat the required closure study. The comparison is the *content*; use matched settings (same problem, same norm, same refinement protocol, stated tolerances) and report the cost measure.

1. **Two boundary closures.** One-sided vs. ghost-point Neumann closure, same code otherwise. Report the observed global order of each and the norm in which the closure's defect is visible.
2. **Direct vs. iterative at matched accuracy.** Sparse direct ('spsolve') vs. Jacobi / Gauss–Seidel / SOR from `iterative_solvers_skeleton.py`. Run the direct solve once; tighten the iterative tolerance until both meet a stated accuracy target (the same scaled residual, or agreement in a quantity of interest). Report iteration counts, wall time, and — if observed — the mesh size at which the crossover occurs; if no crossover occurs over the tested range, report that explicitly. The stopping criterion (residual, not iterate difference) is part of the method.
3. **FDM vs. FEM on the same problem.** If your core direction admits both discretisations (Direction 3 does), run the comparison on your project problem with matched data. Otherwise — the disk and mixed-boundary directions prescribe FDM formulations — this is a separate, reproducible benchmark: solve the 2D Poisson problem on $\Omega = (0,1)^2$ with identical f, Dirichlet data, and manufactured solution, using the five-point FDM stencil on an $N \times N$ square grid and continuous piecewise-linear FEM on the triangulation obtained by splitting those same squares consistently. A 1D FEM calculation is not a valid comparison with the 2D FDM problem; `fem1d_skeleton.py` may be consulted for assembly ideas only, while the comparison itself requires a 2D implementation such as `scikit-fem`. Match the mesh scale h, state both numbers of degrees of freedom, evaluate both errors in the same L^2 norm (and at common nodes for a nodal L^∞ comparison), and use comparable algebraic tolerances. Report assembly time, solve time, nonzeros, error, and observed order separately; do not attribute a difference to the method when it is caused by different PDE data, dimension, mesh scale, norm, or stopping criterion.
4. **Smooth vs. nonsmooth data.** The same solver, one smooth manufactured solution and one that fails regularity (e.g. u_* behaving like $r^{2/3}$ at a re-entrant corner). Report how the observed order degrades and *why*, in the norm you chose.

5. Open directions

1. **L-shaped domain, regularity-aware interpretation.** Use the reproducible singular benchmark
$$\Omega = (-1,1)^2 \setminus [0,1)^2, \qquad -\Delta u = 0 \text{ in } \Omega, \qquad u = g \text{ on } \partial\Omega.$$
Write $(x,y) = (r\cos\theta, r\sin\theta)$ with $\theta \in [\pi/2, 2\pi]$ on the L-shaped domain, set $\phi = \theta - \pi/2 \in [0, 3\pi/2]$, and define
$$u_*(r,\theta) = r^{2/3}\sin\left(\frac{2\phi}{3}\right), \qquad g = u_*|_{\partial\Omega}.$$
This branch convention must be implemented explicitly: compute $\theta = \texttt{atan2}(y,x)$ and add 2π whenever $\theta < \pi/2$. It makes $u_* = 0$ on both boundary rays meeting at the re-entrant corner.

The function is harmonic away from the corner, belongs to $H^{1+s}(\Omega)$ for every $s < 2/3$, but not generally to $H^2(\Omega)$. Do not evaluate its singular gradient at the corner.

For continuous piecewise-linear FEM on quasi-uniform, shape-regular meshes that represent the L-shape exactly, theory therefore predicts, up to an arbitrarily small loss in the exponent,

$$\|\nabla(u_* - u_h)\|_{L^2} = O(h^{2/3}), \qquad \|u_* - u_h\|_{L^2} = O(h^{4/3}).$$

The second estimate is not the usual smooth-domain $O(h^2)$ result: the dual problem has the same limited elliptic regularity, so the Aubin–Nitsche argument supplies only another factor approximately $h^{2/3}$. A nodal L^∞ rate near $2/3$ is a useful empirical diagnostic, not a replacement for these norm-specific estimates. FDM teams may use the aligned five-point grid and discrete analogues of the L^2, energy, and maximum norms, but must label their rates as measured FDM results rather than quoting the FEM argument.

Give at least four uniform refinements and report error against both h and the number of unknowns. For added depth, compare with graded or adaptive refinement concentrated near the corner, using the same error norm and stopping tolerance. For 2D linear FEM, quasi-uniform energy error $O(h^{2/3})$ corresponds to approximately $O(N_{\mathrm{dof}}^{-1/3})$; a well-designed graded/adaptive sequence may recover the optimal $O(N_{\mathrm{dof}}^{-1/2})$ energy-error slope. Measure rather than assume that improvement. Show the meshes, state the marking or grading rule, use sufficiently accurate quadrature for the singular exact solution, and report error, degrees of freedom, nonzeros, solve time, and an estimator or refinement indicator. Verify $f = 0$, the two zero trace rays, and the nonzero outer-boundary data independently before interpreting any convergence plot.

2. **Adaptive refinement.** A simple mark–refine loop (local residual indicator, refine the worst elements) compared against uniform refinement in an error-vs-*degrees-of-freedom* plot. The lesson is that adaptive wins for singular problems and can waste work on smooth ones — measure which.

3. **Perpetual American put (obstacle/free-boundary BVP).** Let $\psi(S) = (K - S)^+$ and, for $S > 0$, define the risk-neutral operator

$$\mathcal{L}V = \tfrac{1}{2}\sigma(S)^2 S^2 V'' + (r - q)SV' - rV, \qquad r > 0, \quad q \geq 0, \quad \sigma(S) > 0.$$

The value is characterised globally by the variational inequality

$$\min\{-\mathcal{L}V,\ V - \psi\} = 0,$$

or equivalently by $-\mathcal{L}V \geq 0$, $V - \psi \geq 0$, and $(-\mathcal{L}V)(V - \psi) = 0$. When the contact set is $[0, S^*]$, $V = K - S$ there and $\mathcal{L}V = 0$ for $S > S^*$; value matching $V(S^*) = K - S^*$ and smooth fit $V'(S^*) = -1$ determine the free boundary. Thus $V = \psi$ is an obstacle/contact condition on a region, not a boundary condition imposed only at S^*.

Stage 1—exact validation case. Take $K = 100$, $r = 0.05$, $q = 0$, and constant $\sigma = 0.20$. The decaying root of $\frac{1}{2}\sigma^2\lambda(\lambda - 1) + r\lambda - r = 0$ is $\lambda_- = -2r/\sigma^2 = -2.5$, and

$$S^* = \frac{\lambda_- K}{\lambda_- - 1} = 71.428571\ldots, \qquad V(S) = \begin{cases} K - S, & 0 \leq S \leq S^*, \\ (K - S^*)\big(S/S^*\big)^{\lambda_-}, & S > S^*. \end{cases}$$

Use this case to validate both the computed value and the recovered free boundary; smooth fit is an additional diagnostic, not the sole validation test.

Stage 2—a genuinely numerical local-volatility case. Keep $K = 100$, $r = 0.05$, and set $q = 0.02$ and

$$\sigma(S) = 0.20\left[1 + 0.35\tanh\left(\frac{\log(S/K)}{0.40}\right)\right].$$

This positive, variable coefficient destroys the Euler–Cauchy closed form. Truncate to $[0, S_{\max}]$, impose $V(0) = K$ and $V(S_{\max}) = 0$ as the far-field approximation, and repeat with (for example)

$S_{\max}/K = 3, 4, 6$ so that truncation error is not confused with mesh error. Discretise the obstacle problem as a linear complementarity problem (LCP), not as one linear solve. Near $S = 0$ the operator is degenerate and centred drift differencing need not be monotone: use an upwind/fitted drift treatment, or demonstrate that your chosen stencil gives an M-matrix (check the off-diagonal signs and discrete maximum principle).

Solve the LCP by projected SOR or policy iteration. State the initial iterate, relaxation parameter (if used), iteration count, and stopping rule. With $a_h = -\mathcal{L}_h V_h$ and $b_h = V_h - \psi_h$, report

$$\min_i(a_{h,i}), \qquad \min_i(b_{h,i}), \qquad \|\min(a_h, b_h)\|_\infty,$$

where the last minimum is componentwise. Also verify $\psi_h \le V_h \le K$, monotonic decrease and convexity of the put value, and a single contiguous contact set from $S = 0$.

Define explicitly how S_h^* is extracted from the contact-to- continuation transition (including its tolerance and any interpolation). Give at least four mesh refinements. For Stage 1, tabulate $\|V_h - V\|_\infty$ and $|S_h^* - S^*|$; for Stage 2, use a refined-grid reference and separate the h-study from the $S_{\max}$-study. A two-asset perpetual American spread option is an optional ambitious 2D extension, but it requires a separately stated stochastic model, payoff, computational domain, and boundary conditions; it is not the same numerical problem as this 1D LCP.

4. **Electrostatic MEMS pull-in (nonlinear, with bifurcation).** The voltage-driven membrane of the previous course material,

$$-u''(x) = \frac{\lambda}{[1 - u(x)]^2}, \quad -1 < x < 1, \qquad u(\pm 1) = 0, \qquad 0 \le u < 1.$$

Below a critical λ^* two static equilibria coexist; at λ^* they merge in a fold; above it none exists. The content is the derivation and the judgment: nondimensionalise the force balance; derive the exact branch parametrisation $\lambda(a)$ from a first integral; implement finite differences with a feasibility-preserving damped Newton (`mems_skeleton.py`); follow the branch by continuation; locate the fold with a defensible error bar; cross-check against an independent solver (e.g. `scipy.integrate.solve_bvp`). "Newton converged" is not "the problem is solved" (two branches exist — did you find both?), and Newton *failure* near the fold has at least three competing explanations you must distinguish experimentally. A linear warm-up (steady heat fin, $-u'' + m^2 u = f$ with a closed-form solution) builds the stencil and the evidence habits first. A 2D membrane extension ($-\Delta u = \lambda/(1-u)^2$ on a rectangle) is the ambitious variant.

5. **Microstrip transmission line (2D, signal integrity).** Model the cross-section of a printed-circuit-board microstrip as a 2D electrostatic BVP with spatially varying permittivity,

$$-\nabla \cdot \big(\epsilon(x, y)\, \nabla\phi\big) = 0 \quad \text{in } \Omega_R, \qquad \epsilon = \epsilon_0 \epsilon_r(x, y),$$

with $\phi = V_0$ on the signal trace, $\phi = 0$ on the ground plane, and a stated far-field condition (homogeneous Neumann or Dirichlet) on the truncated boundary Γ_{far}; across the air–substrate interface the flux continuity $[\epsilon \nabla\phi \cdot n] = 0$ must hold. This is the divergence-form variable-diffusion operator of core direction #3 with a *discontinuous* coefficient and real geometry, so FEM is the natural choice and the weak form enforces interface flux continuity in the distributional sense, without Lagrange multipliers. The content is the engineering loop: pick one far-field condition and a reproducible domain-growth path, study the truncation bias; resolve the conductor re-entrant-corner singularity (cf. the L-shaped direction) with graded or adaptive refinement; then extract the capacitance per unit length C, effective permittivity $\varepsilon_{\text{eff}} = C/C_{\text{air}}$, and characteristic impedance $Z_0 = \sqrt{\varepsilon_{\text{eff}}}/(c_0 C)$ and cross-check against the IPC-2141 and Hammerstad–Jensen empirical formulas. A 2D FEM library (`scikit-fem` or FEniCS) supplies the weak form, function space, and boundary treatment; the discipline is unchanged from the core directions — matched settings, a named norm, and a refinement table whose observed order is declared *before* you claim it. More details can be found in `Capacitor2d.pdf`.

6. Standing protocols (all directions)

- **Algebraic-error separation.** On at least one representative mesh, show that the algebraic error is negligible relative to the discretisation error. Iterative-solver teams: repeat the solve at successively tighter tolerances and confirm that the reported quantity stops changing. Direct-solver (`spsolve`) teams: `spsolve` has no tolerance parameter — report its backward (residual) error and bound the forward error with a conditioning estimate, or cross-check the solve independently at a very tight tolerance. Report a scaled or backward residual separately; residual size is not solution error without information about the inverse operator and conditioning.
- **Questions before calculation.** Before each new solve, complete the pre-computation protocol (`precomputation_questions_worksheet.tex`): written predictions, then reconciliation. A prediction edited to match output is a falsified record.
- **AI audit.** Any AI-generated solver code must pass the five-gate audit (`ai_solver_audit_worksheet.tex`) before its output is used, and the audit record attaches to the AI Transparency Log.

Choosing your direction

Consideration	Suggests
You want the first taste of a curved boundary	#1 (disk MMS)
You want real boundary-condition content	#2 (mixed data)
You want a physical heterogeneous medium	#3 (variable diffusion)
You want to compare methods	a depth direction on #1 or #2
You like singular/nonlinear mathematics	L-shaped, adaptive, or MEMS
You like finance	perpetual American option
You like electromagnetics / a real engineering validation	microstrip transmission line

Whichever you choose, use the scaffolding as a starting point rather than as a completed solver. The test-problem bank supplies analytic data and checks the manufactured forcing for the rectangular benchmarks; it does not verify a student implementation. The Poisson and iterative-solver files deliberately leave matrix assembly, convergence studies, and solver sweeps as `TODO`s. The convergence template supplies plotting and order-fitting utilities once you provide computed errors. Likewise, the MEMS scaffold provides the residual, Jacobian audit, and experiment driver, but students must implement the Jacobian, damped Newton method, and continuation. The disk and American-option directions require their own cut-cell and LCP implementations, respectively. Stars classify scope: the higher levels open up the depth-and-novelty space scored in Dimension B. Within a chosen scope, quality and verification decide the grade — excellent, thoroughly verified #1 work can earn a strong mark.